xForge V2

Provider-Agnostic Football Analytics Pipeline

End-to-end engineering case study

The Problem

Traditional football analytics pipelines are locked to a single data provider. When clubs change providers or work with multiple feeds simultaneously, data engineers must rewrite ingestion, coordinate systems, and downstream models from scratch. Coordinates differ: StatsBomb uses 120x80, Opta uses 100x100, real pitches measure 105x68 m. There is no industry standard.

The Solution — Architecture

xForge V2 is built on three load-bearing design principles:

- **Adapter Pattern (Dart microservice):** StatsBombAdapter and OptaAdapter both implement a DataAdapter interface. Adding a third provider requires one new file — zero changes to downstream models.
- **Universal Coordinate System:** a single dbt macro (normalise_x / normalise_y) converts any provider's raw coordinates to 105x68 m on first touch in the Silver layer. Spatial range tests gate the pipeline on every run.
- **Medallion Architecture:** Bronze (raw + type-cast) → Silver (normalised + tested) → Gold (aggregated) → Marts (BI-ready). ML models write xG / xT / xP values back to fact_events before Gold materialises.

StatsBomb / Opta	Dart HTTP :8090	PostgreSQL 15	dbt Medallion	XGBoost / xT	BI / XML / PDF
Open data feeds	Adapter pattern	LIST-partitioned	4-layer pipeline	Calibrated models	Superset + Hudl

Technology stack: Dart (AOT-compiled microservice), Python 3.11, PostgreSQL 15 (LIST-partitioned), dbt-core, XGBoost, scikit-learn, Apache Airflow, GitHub Actions CI, Apache Superset.

147	524,457	118,187	0.897	~11 min	5
WC 2022 matches	Events ingested	xP training passes	xP model AUC	Full CI runtime	BI chart artifacts

ML Models

Model	Method	Key Result
xT Surface	Value iteration, 15 passes, 16x12 grid, 192 cells	Max xT = 0.298 (penalty area)
xG Classifier	XGBoost + Platt scaling (CalibratedClassifierCV)	Calibrated probability — xG=0.30 represents ~30% conversion
xP Classifier	XGBoost + chunked write-back	AUC 0.897, log-loss 0.293, trained on 118,187 passes
Set-Piece Clustering	K-Means k=6 on 105x68 m coordinates	12 centroids across 1,384 corners and 4,904 shots
Press Trigger Detector	Rule-based: ball recovery + 3 defensive actions / 5 s	165 triggers detected across 147 WC 2022 matches

Validation — WC 2022 (147 Matches, 524,457 Events)

Every push to main triggers a full end-to-end pipeline run on GitHub Actions:

- 147 World Cup 2022 matches ingested via the Dart HTTP API
- dbt Silver spatial range tests act as a hard gate — the pipeline fails if any coordinate falls outside 105x68 m bounds after normalisation
- All five ML models retrained and validated on each run
- Five BI charts generated and uploaded as 90-day artifacts (xT heatmap, shot map, team xG comparison, set-piece clusters, player xP ranking)
- Full pipeline duration: approximately 11 minutes

Business Applications

- **Pass Quality Scouting** — xP isolates pass difficulty from completion rate. High-difficulty completions in high-xT zones surface midfielders invisible to traditional completion-rate leaderboards.
- **Set-Piece Intelligence** — K-Means delivery zones across 147 matches replace manual video tagging for opponent set-piece preparation.
- **Video Integration** — SportsCode/Hudl-compatible XML with the top-25 xT events per match. Analysts open the file directly in Hudl Sportscode — no manual timestamp entry.
- **Recruitment** — finishing_quality = goals – total_xG separates clinical finishers from shot-volume players; directly queryable in mart_player_metrics.

github.com/bbasaranemir/xforge CI: all runs passing on main kircaali036@gmail.com